

Hexagonal Cedar Timber Pergola — Master Construction Plan

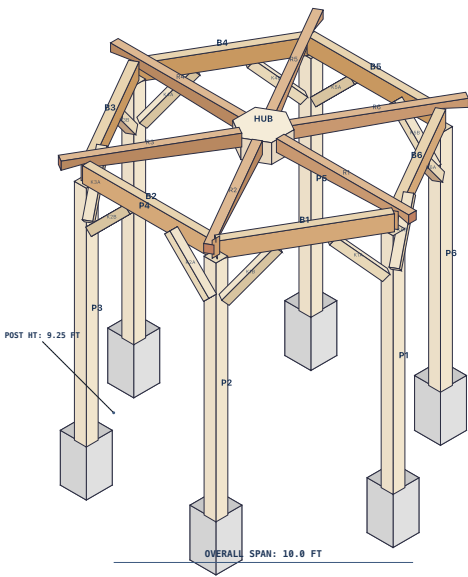
Location: Saanich, BC
Style: Warm cedar timber — open-frame, mortise & tenon, no metal hangers
Visual Intent: Heavy 6×6 warm cedar posts on visible concrete footing pads; exposed 4×6 hip rafters with decorative tails; heavy 6×12 beam ring; diagonal knee braces; traditional mortise-and-tenon joinery.
Revision: 1.3.1 | August 2026

1. Architectural 3D & Elevation Views

3D Photorealistic Render

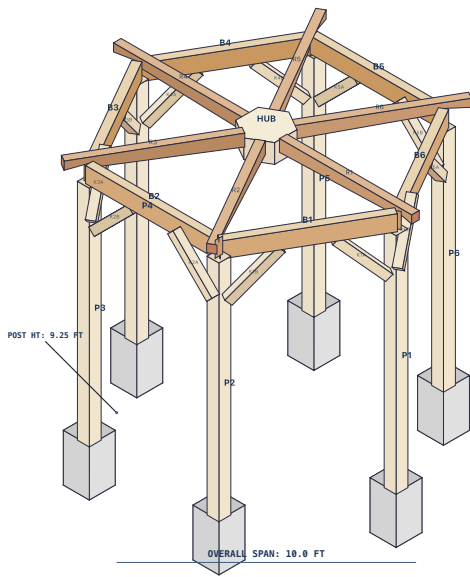


Perspective View



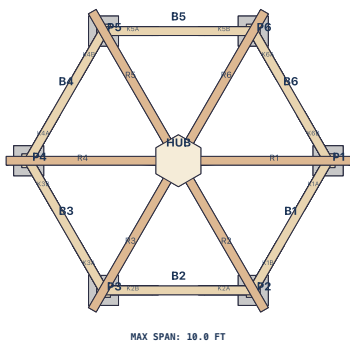
DRAWING PERSPECTIVE VIEW
STRUCTURE: PERGOLA JURISDICTION: BC_COASTAL SOURCE HASH: a1b2c3d4 DATE: 2026-05-24 SCALE: AUTO

Isometric View



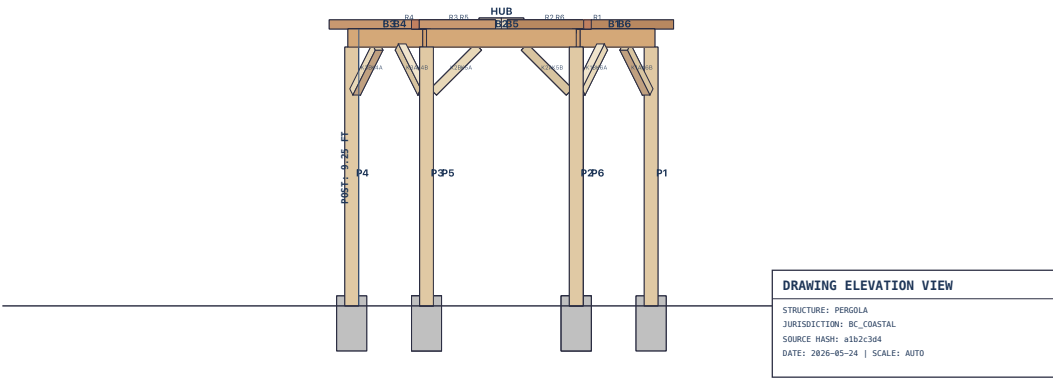
DRAWING ISOMETRIC VIEW
STRUCTURE: PERGOLA JURISDICTION: BC_COASTAL SOURCE HASH: a1b2c3d4 DATE: 2026-05-24 SCALE: AUTO

Plan View



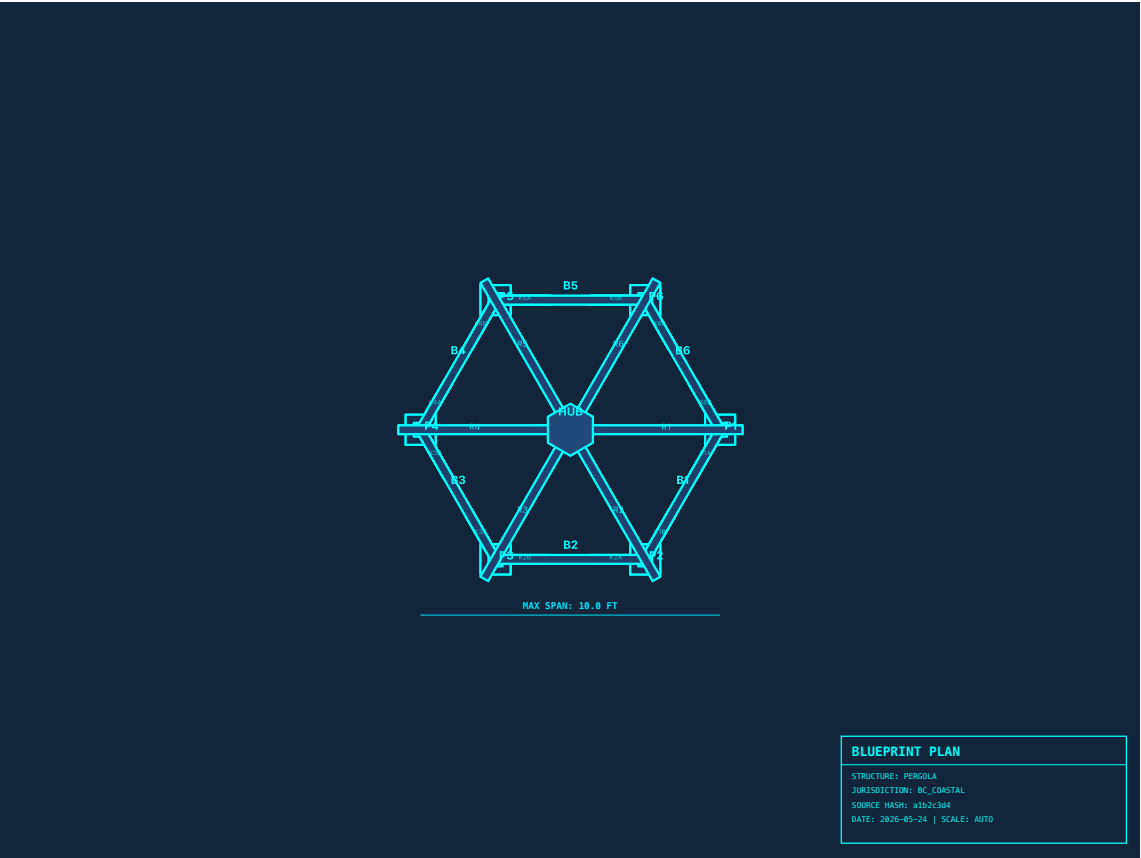
DRAWING PLAN VIEW
STRUCTURE: PERGOLA JURISDICTION: BC_COASTAL SOURCE HASH: a1b2c3d4 DATE: 2026-05-24 SCALE: AUTO

Elevation View

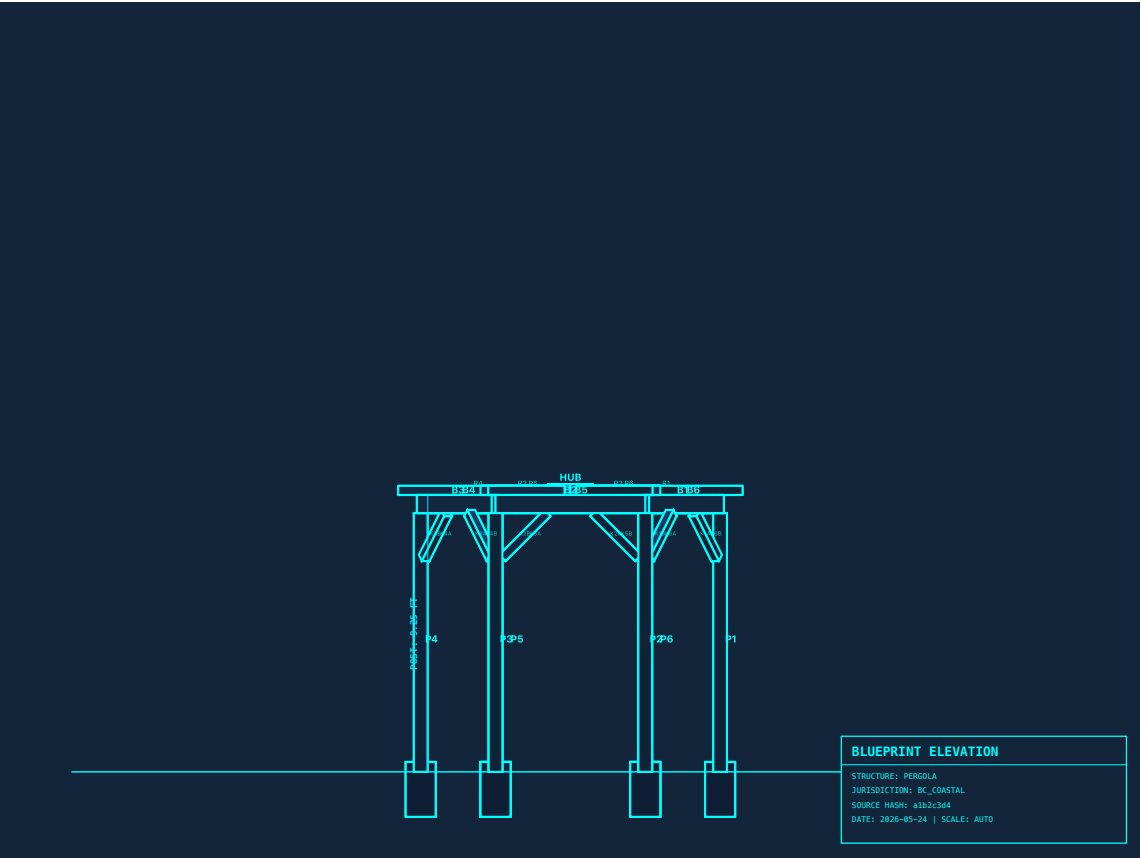


2. Technical Blueprints & Component Details

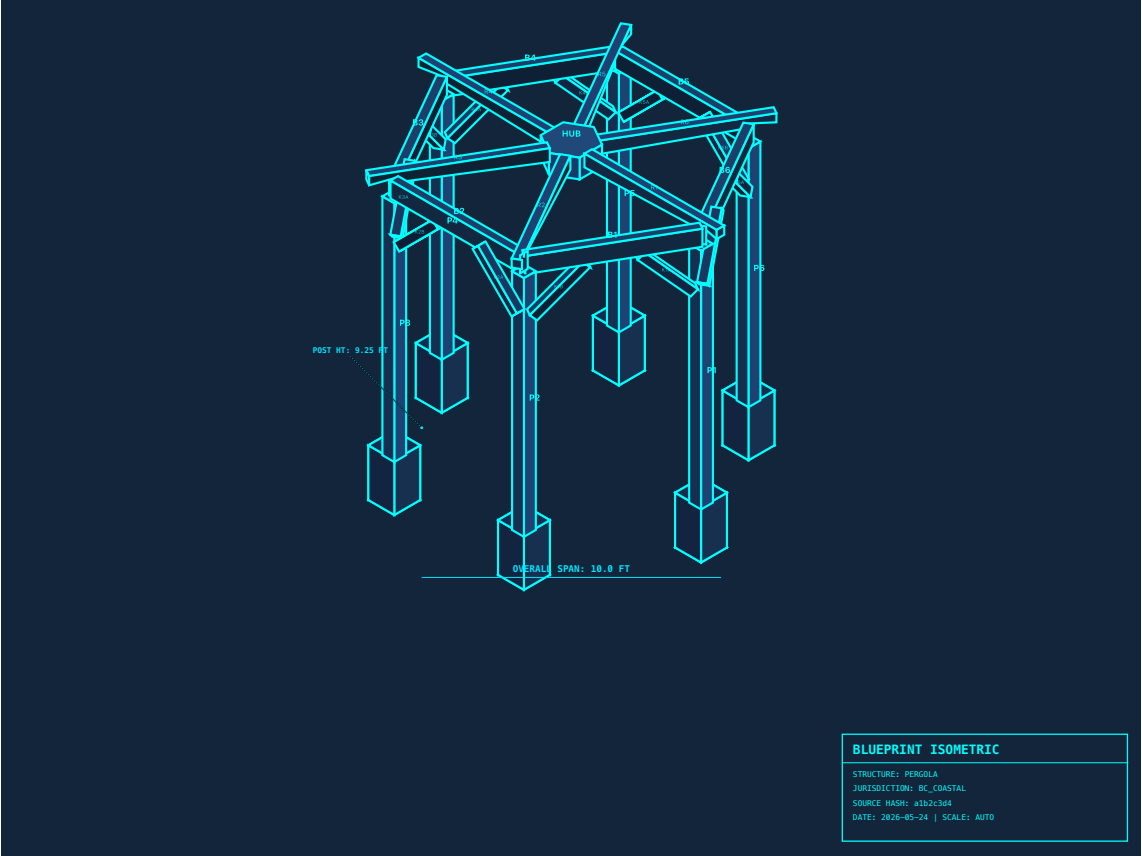
SB-01: Blueprint Plan View



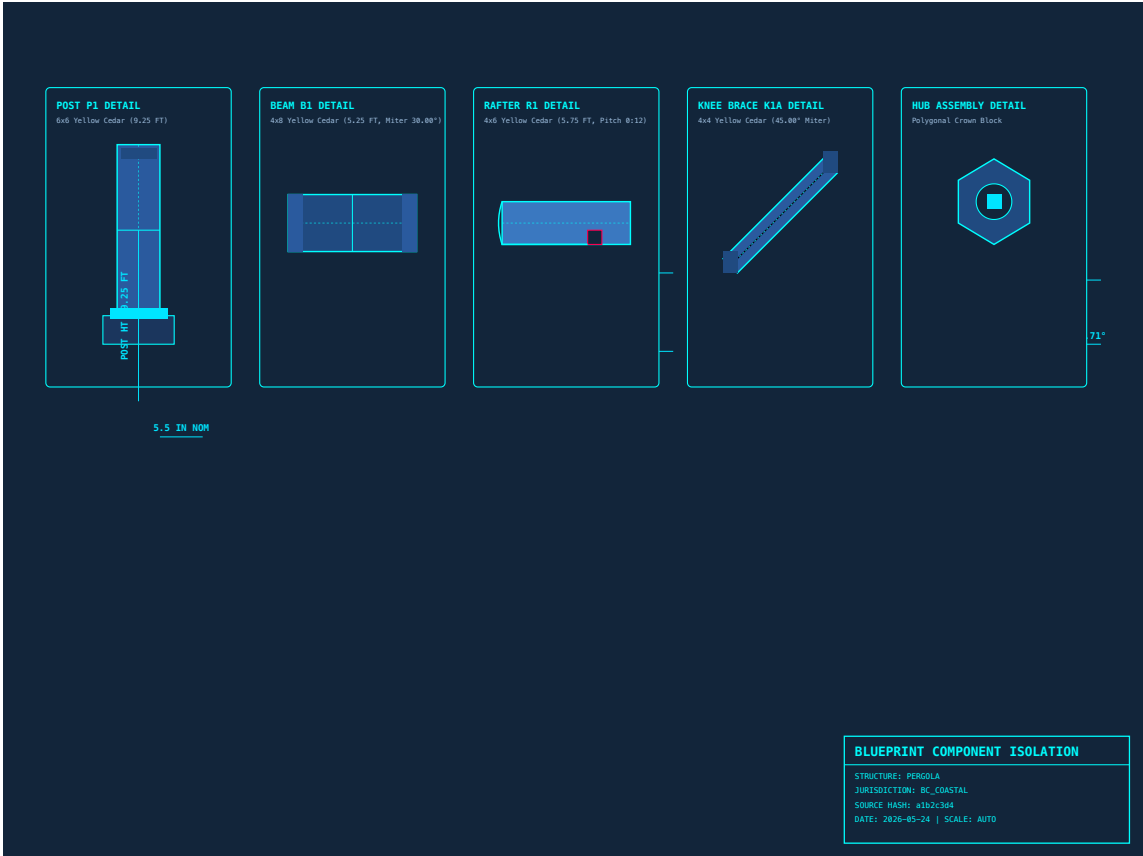
SB-02: Blueprint Elevation View



SB-03: Blueprint Isometric View



SB-04: Component Isolation Details



3. Structural & Geometric Summary

Parameter	Specification	Source
Footprint Radius	4.875 ft (Inscribed Hexagon)	structure.json
Post Size & Height	6x6 (5.5"x5.5") @ 7.42 ft soffit	structure.json
Beam Size & Span	6x12 (6.0"x12.0") @ 4.875 ft span	structure.json
Rafter Size & Pitch	4x6 (3.5"x5.5") @ 4:12 pitch	structure.json
Rafter Miter / Bevel	28.71° Miter / 9.10° Bevel	geometry_engine.py
Beam Ring Miter	30.00° Miter	geometry_engine.py
Total Height	11.05 ft	geometry_engine.py
Snow / Wind Load	40 psf snow / 25 psf wind	building-code.json